Six Sigma in the Aerospace Industry: A Case Study Review

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Abstract

This aim of this paper is to investigate and discuss applications of Six Sigma in the Aerospace Industry. Beginning with an in-depth review of the aerospace industry and brief introduction to Six Sigma, the tools and methodology of implementation of Six Sigma stages are studied with a particular focus on the process of 'Define, Measure, Analyze, Improve and Control cycle' (DMAIC). An example case study of the focused methodology is discussed. This study is discussed in detail to provide additional comprehension as to how the aforementioned methods may be applied to a practical application and how they may benefit processes seeking reduction in variability and an increase in product quality.

Keywords
Six Sigma, Process Engineering, Aerospace Industry, DMAIC, Lean Manufacturing

1. Introduction
Most organizations on the global market are well aware of the importance of product quality and continuous improvement of quality and efficiency and the aerospace industry is no different [1]. According to a recent estimation, the global aerospace and defense industry is worth a massive US$685.6 billion [2]. With an industry of this size, competition for research and development, along with the struggle to remain efficient and minimize costs is a constant concern. The economic life of an aircraft is estimated to be an average of 25 years and must undergo numerous customary checks as well as maintenance to be considered to be in operative condition [3]. Beyond these requirements, older aircrafts are often refurbished and overhauled under manufacturer specifications. With such detailed specifications, customary checks, and maintenance, remaining relevant in the aerospace industry requires companies to produce high quality aircrafts and keep up with all standards. In addition to this, due to the fact that passengers will be on board, safety is a top concern. To increase competitiveness, maintain quality, and ensure efficiency, aerospace companies use several statistical, executive, and engineering techniques. One of these methods is Six Sigma (SS), which was initiated by Motorola in 1987 in order to eliminate variations in the process and provide significant improvement to the product and overall process [4]. Six Sigma is a statistical concept, which measures a process in terms of Defects per Million Opportunities (DPMO) and equates to 3.4 DPMO. This allows for companies and manufacturers to aim towards a measurement, which minimizes defects generated by their processes. Another main focus of Six Sigma is to reduce variability and therefore have a process that is more in control. One approach to reduce the inherent variability in manufacturing is known as DMAIC, which is comprised of the following phases:

- Define
- Measure
- Analyze
- Improve
- Control

Using the DMAIC methodology allows one to strategically look at their process in a step-by-step manner to ensure the right issues are being addressed, improved, and positive results are sustained. Beginning with the define stage, the problem is defined along with a summary of costs, benefits, and impact on the customer [5]. Following the task of defining the problem at hand and project scope, the measure phase ensues and operational definitions for each critical-to-quality (CTQ) characteristic are developed; the scope of all parameters as well as their performances are established. Next, the analyze phase is completed by identifying key causes and process determinants; if data from

the current process exists, it will be explored in the analyze phase to further investigate where the problems within the process lie [6]. The improve step involves using all previous steps to change the process and optimize performance through reducing variability and directly addressing main root causes with solutions. Finally, the control phase is used to sustain positive changes and strive for continuous improvement. As the individual who is trying to improve their process moves through the steps of DMAIC, it is helpful to ask questions such as those seen in Figure 1 below [6].

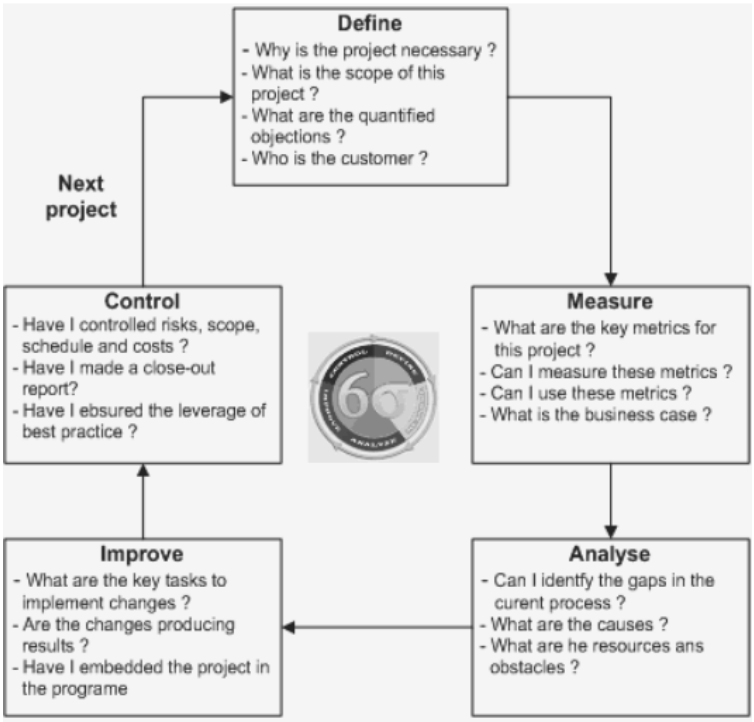

Figure 1: The DMAIC cycle as a methodology of a Six Sigma

To further supplement the questions seen in Figure 1, there are many tools that can help reaching viable conclusions within each phase. A few of these tools will be explained in detail within the discussion of the case study, but various tools which may be used in each phase of DMAIC are shown in Figure 2 [7].

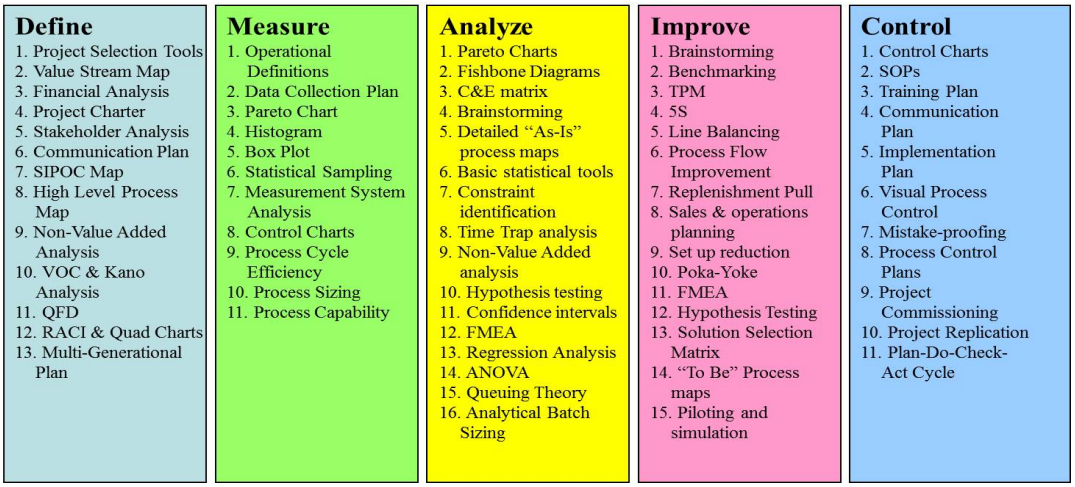

Figure 2. Tools Used in Each Phase of DMAIC

2. Case Study of Six Sigma

The case study discussed involves a family owned company which offers machining services to the aerospace industry and desires to reduce the cost of manufacturing by the reduction of machining cycle time [8]. With the wish to reduce variability in production and reduce overall cycle time, the DMAIC methodology was applied to the process. Reference [9] describes how the company exposed themselves to Six Sigma and the main Project Engineer decided to make use of DMAIC enhancement procedure in order to decrease cycle time. The details of what tools were used at each phase of DMAIC and how these steps aided in the reduction of cycle time are discussed in further detail in sections 2.1-2.5.

2.1 Define

In the define phase, it is important to establish the goals of the project and the needs of the customer. Prior to the case study, the cycle time was over 1200 seconds and had a standard deviation of 31 seconds [9]. The Project Engineer decided that the target cycle time reduction of 50%; it was determined that this reduction would not negatively affect the variation. Additional goals were set such as reduction in overtime, and reduced allotment time for assessment.

2.2 Measure

The measure phase of the DMAIC process determines the current performance metrics and assesses the best way to use said metrics. For this particular case study, input, process, and output metrics are classified and specified. From those metrics determined, 9 were considered to be key metrics and further investigated. Operational descriptions were explained in detail and new data collection techniques were established. Gauge Reproducibility and Repeatability was also conducted on the fresh system of measurement to ensure discrepancy of the study and results showed less than five percent. Further, data was explored to ensure stability and also illustrated no presence of special cause variation. Finally, the sigma value was measured for the process and the result was a large negative value.

2.3 Analyze

The analyze phase requires an evaluation and definition of the cause of defects and variation. The data, which was obtained in the measure phase, is now analyzed to investigate the cause of higher cycle times during production. While authenticating four considerable factors as root causes for the increased cycle time, evaluating the proficiency of the team, and conducting extensive statistical investigation, it was found that one of the considerable factors caused over 40% of the inherent variability and overall increase in production time.

2.4 Improve

The improve phase looks to eradicate defects and enhance the process. In this phase, action is taken to eliminate or help alleviate factors, which have negative effects on the overall process, per determined by the analyses in the previous phase. The case study notes that there were prospective illuminations that were produced by the solution generation practices as well as models of lean [9]. The elucidations were used to convince the organization that they would lead to promising results and they were ultimately put into action by the organization. Validation through statistical means was used to demonstrate the viability of the proposed solutions and found that the new solution returned a sigma value in excess of five. Further, control charts were utilized to illustrate the stability of the new procedure and were found to be in statistical control. Overall, a 20% reduction in the number of steps throughout the procedure resulted after implementing suggested solutions.

2.5 Control

Performance of the future procedures must be sustained and controlled; in this phase of DMAIC, a plan must be put in place, which will prevent the process from returning to its previous detrimental ways. To maintain the positive results found through the improve phase, standardization was use. All old documents were updated and a new procedure map was put in place. In addition to this documentation, training plans were developed and implemented to ensure all employees were on board with the new procedure. Potential future problems were predicted so a preventative plan may be put in place. The potential problems were classified into three distinct categories, which were techniques, environment, and human factors. Following the identification and classification of these potential difficulties, control plans were developed and put in place to not only monitor and supervise, but also take further action when necessary. Finally, lessons learnt as well as overall outcomes were calculated to further solidify the

benefits of implementation and achievement of overall goal in reduction of cycle time. The original time against the improvement time is shown below in Table 1 [9].

Table 1: Original Time Against Improvement Time

Description	Original	New
Average Cycle Time (seconds)	1245	670
Standard Deviation (seconds)	31	5.5
Percentage Improvement (%)	N/A	46

3. Discussion and Conclusions

As apparent from Table 1, the use of the DMAIC process and Six Sigma tools helped to improve the cycle time by nearly 50%. The standard deviation was decreased significantly and there was an overall percentage improvement of 46. Due to the fact that the cycle time was reduced to such a significant extent, a large cost reduction in the process has also occurred.

This is just one example of hundreds of cases in which Six Sigma methodology and the DMAIC process were introduced to a process to reduce costs and variability while increasing product quality [10]. The beauty of DMAIC is that it may be applied to any process seeking a reduction in variability and an overall reduction in 'defects', whatever those defects may be classified as. Because this approach uses a step-by-step methodology, it ensures the analyst is looking at every aspect of the process and finding the true root causes and working to efficiently eradicate them. Having the ability to analyze the process as a whole, and focus on the tasks, which are most effective in improving your process, is incredibly valuable. For industries such as global aerospace and defense, using approaches such as these are vital to maintain relevant, competitive, and efficient.

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